

Data Cleaning: Eurovision

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Rule 1: Look at the Data

Rule 2: Is There a Package to Deal With That

```
songs <- jsonlite::fromJSON(glue::glue("./data/eurovision-lyrics.json"))
```

Rule 3: Break it Into Pieces

```
songs[[1]]
```

```
## $'#'  
## [1] "1"  
##  
## $Country  
## [1] "Netherlands"  
##  
## $#.1'  
## [1] "1"  
##  
## $Artist  
## [1] "Jetty Paerl"  
##  
## $Song  
## [1] "De vogels van Holland"  
##  
## $Language  
## [1] "Dutch"  
##  
## $Pl.  
## [1] "-"  
##  
## $Sc.  
## [1] "-"  
##  
## $Eurovision_Number  
## [1] 1  
##  
## $Year  
## [1] "1956"  
##  
## $Host_Country  
## [1] "Switzerland"  
##  
## $Host_City  
## [1] "Lugano"  
##  
## $Lyrics  
## [1] "De vogels van Holland zijn zo muzikaal\nZe leren in hun prille jeugd al tierelieren\nDe merel, o  
##  
## $Lyrics translation'  
## [1] "The birds of Holland are so musical\nThey already learn to twitter in their early youth\nThe bl
```

```

parse_song <- function(song){
  country <- song$Country
  artist <- song$Artist
  title <- song$Song
  year <- song$Year
  lyrics <- song$Lyrics
  translation <- song$`Lyrics translation`
  place <- song$Pl.
  tibble(
    country, artist, title, year, lyrics, translation, place
  )
}
parse_song(songs[[1]])

```

```

## # A tibble: 1 x 7
##   country      artist      title      year lyrics  translation place
##   <chr>      <chr>      <chr>      <chr> <chr>   <chr>      <chr>
## 1 Netherlands Jetty Paerl De vogels van Holland 1956 "De vog~ "The birds~ -

```

Rule 4: Get it Into a Tibble

```

songs <-
  songs %>%
  map_df(parse_song, .id = "ID")
songs

```

```

## # A tibble: 1,644 x 8
##   ID      country      artist      title year lyrics  translation place
##   <chr> <chr>      <chr>      <chr> <chr> <chr>   <chr>      <chr>
## 1 0      Netherlands Jetty Paerl De v~ 1956 "De v~ "The birds~ -
## 2 1      Switzerland Lys Assia  Das ~ 1956 "Das ~ "The old c~ -
## 3 2      Belgium      Fud Leclerc Mess~ 1956 "Mess~ "Ye drowne~ -
## 4 3      Germany (West) Walter Andreas Sch~ Im W~ 1956 "Es g~ "There is ~ -
## 5 4      France        Mathé Altéry Le t~ 1956 "Chan~ "Sing, car~ -
## 6 5      Luxembourg    Michèle Arnaud Ne c~ 1956 "Si o~ "If they t~ -
## 7 6      Italy          Franca Raimondi Apri~ 1956 "La p~ "The first~ -
## 8 7      Netherlands Corry Brokken Voor~ 1956 "Voor~ "Over for ~ -
## 9 8      Switzerland Lys Assia (2) Refr~ 1956 "[Int~ "Chorus of~ 1
## 10 9     Belgium      Mony Marc  Le p~ 1956 "Les ~ "The bells~ -
## # i 1,634 more rows

```

Rule 5: Look at Each Column

We need to make sure all of our data makes sense in context. You can find the source data at this link: <https://www.kaggle.com/minitree/eurovision-song-lyrics>

ID

This is an ID number assigned to each song. We should check that the song IDs are all unique.

```
songs %>% count(ID) %>% filter(n > 1)
```

```
## # A tibble: 0 x 2
## # i 2 variables: ID <chr>, n <int>
```

They are unique, but they're also not useful, so delete them.

```
songs <-
  songs %>%
    select(-ID)
songs
```

```
## # A tibble: 1,644 x 7
##   country      artist      title    year lyrics translation place
##   <chr>        <chr>      <chr>    <chr> <chr> <chr>      <chr>
## 1 Netherlands Jetty Paerl    De voge~ 1956 "De v~ "The birds~ -
## 2 Switzerland Lys Assia     Das alt~ 1956 "Das ~ "The old c~ -
## 3 Belgium      Fud Leclerc   Messieu~ 1956 "Mess~ "Ye drowne~ -
## 4 Germany (West) Walter Andreas Schwarz Im Wart~ 1956 "Es g~ "There is ~ -
## 5 France        Mathé Altéry  Le temp~ 1956 "Chan~ "Sing, car~ -
## 6 Luxembourg    Michèle Arnaud Ne croi~ 1956 "Si o~ "If they t~ -
## 7 Italy          Franca Raimondi Aprite ~ 1956 "La p~ "The first~ -
## 8 Netherlands Corry Brokken Voorgoe~ 1956 "Voor~ "Over for ~ -
## 9 Switzerland Lys Assia (2) Refrain  1956 "[Int~ "Chorus of~ 1
## 10 Belgium      Mony Marc    Le plus~ 1956 "Les ~ "The bells~ -
## # i 1,634 more rows
```

Artist

Are there any artists who appear more than once?

```
songs %>%
  count(artist) %>%
  arrange(-n)
```

```
## # A tibble: 1,621 x 2
##   artist      n
##   <chr>    <int>
## 1 Ana Soklič      2
## 2 Benny Cristo    2
## 3 Blas Cantó      2
## 4 Daði og Gagnamagnið 2
## 5 Destiny         2
## 6 Eden Alene       2
## 7 Efendi           2
## 8 Gjon's Tears     2
## 9 Go_A             2
## 10 Hooverphonic     2
## # i 1,611 more rows
```

```
songs %>%
  filter(str_detect(artist, "Jed"))
```

```
## # A tibble: 2 x 7
##   country artist      title      year lyrics      translation place
##   <chr>   <chr>      <chr>    <chr> <chr>      <chr>    <chr>
## 1 Ireland Jedward    Lipstick  2011  "You say you're on it b~ English    8
## 2 Ireland Jedward (2) Waterline 2012  "Floodgates, can't wait~ English    19
```

```
songs %>%
  filter(country == "Australia")
```

```
## # A tibble: 7 x 7
##   country artist      title      year lyrics      translation place
##   <chr>   <chr>      <chr>    <chr> <chr>      <chr>    <chr>
## 1 Australia Guy Sebastian Tonight Again  2015 "[Verse~ English    5
## 2 Australia Dami Im      Sound of Silence 2016 "[Verse~ English    2
## 3 Australia Isaiah      Don't Come Easy  2017 "[Verse~ English    9
## 4 Australia Jessica Mauboy We Got Love      2018 "[Verse~ English   20
## 5 Australia Kate Miller-Heidke Zero Gravity      2019 "[Verse~ English    9
## 6 Australia Montaigne    Don't Break Me   2020 "I don'~ English    -
## 7 Australia Montaigne    Technicolour     2021 "I wann~ English    -
```

Title

```
songs %>%
  count(title) %>%
  arrange(-n)
```

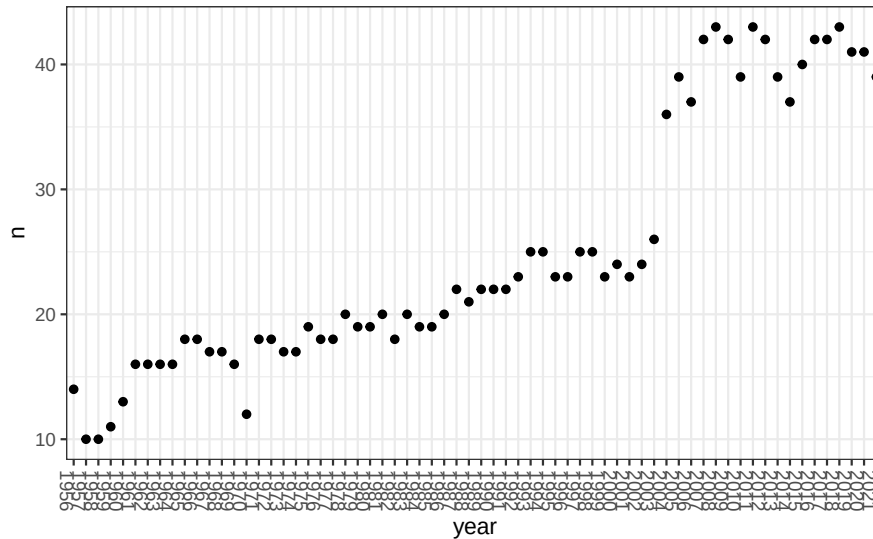
```
## # A tibble: 1,608 x 2
##   title      n
##   <chr>    <int>
## 1 Shine      4
## 2 Angel      3
## 3 Time       3
## 4 Amen       2
## 5 Attention  2
## 6 Casanova  2
## 7 Coming Home 2
## 8 Congratulations 2
## 9 Fairytale  2
## 10 Goodbye   2
## # i 1,598 more rows
```

Year

```
range(songs$year)
```

```
## [1] "1956" "2021"
```

```
songs %>%  
  count(year) %>%  
  ggplot(aes(year, n)) +  
  geom_point() +  
  theme(axis.text.x = element_text(angle = -90, hjust=0))
```



Lyrics

```
songs %>%  
  sample_n(5) %>%  
  pull(lyrics) %>%  
  cat(sep = "\n\n\n")
```

```
## Jednom kad zaboraviš mi ime  
## I krevet hladan postane  
## Kao snijegom okovane zime  
## Kad ništa ne preostane  
## Ja ću te voljeti i tad  
## Divlji vjetre  
## U tišini ostavi me gdje sam  
## Svaka riječ sad prosta je  
## Odlazi i pusti me da ne znam  
## Kako čovjek postaje  
## Ono što nisi više  
## Divlji vjetre  
## Tu na moj prag  
## Uvelo lišće donesi mi  
## I kišu neka ostavi trag  
## Tuge u bojama jeseni  
## A činilo se  
## Zauvijek smo dušom i tijelom zaneseni
```

```

## Ostaje mi
## Tuga u bojama jeseni
## Ono što nisi više...
## Hladna kao
## Kiša ledena
## Kada voljet prestaneš
## I kidaj grane
## Nukrotivo
## Ali prije nego nestaneš
## Divlji vjetre
## Tu na moj prag
## Uvelo lišće donesi mi
## I kišu neka ostavi trag
## Tuge u bojama jeseni
## A činilo se
## Zauvijek smo dušom i tijelom zaneseni
## Divlji vjetre
## Ostaje mi
## Tuga u bojama jeseni
## Tuga u bojama jeseni
## Tuga u bojama jeseni
##
##
## [Intro]
## Hard rock, hallelujah!
## Hard rock, hallelujah!
## Hard rock, hallelujah!
## Hard rock, hallelujah!
##
## [Verse 1]
## The saints are crippled on this sinners' night
## Lost are the lambs with no guiding light
## The walls come down like thunder, the rock's about to roll
## It's the arockalypse, now bare your soul
##
## [Pre-Chorus]
## All we need is lightning with power and might
## Striking down the prophets of false
## As the moon is rising, give us the sign
## Now let us rise up in awe
##
## [Chorus]
## Rock 'n roll angels bring thyn hard rock, hallelujah
## Demons and angels, all in one have arrived
## Rock 'n roll angels bring thyn hard rock, hallelujah
## In God's creation, supernatural high
##
## [Verse 2]
## The true believers thou shall be saved
## Brothers and sisters, keep strong in the faith
## On the day of Rockoning it's who dares, wins
## You will see the jokers soon'll be the new kings
##
## [Pre-Chorus]

```

```

## All we need is lightning with power and might
## Striking down the prophets of false
## As the moon is rising, give us the sign
## Now let us rise up in awe
##
## [Chorus]
## Rock 'n roll angels bring thyn hard rock, hallelujah
## Demons and angels, all in one have arrived
## Rock 'n roll angels bring thyn hard rock, hallelujah
## In God's creation, supernatural high
##
## [Bridge]
## Wings on my back, I got horns on my head
## My fangs are sharp and my eyes are red
## Not quite an angel or the one that fell
## Now choose to join us or go straight to Hell
## Hard rock, hallelujah!
## Hard rock, hallelujah!
## Hard rock hallelujah!
## Hard rock, yeah!
##
## [Chorus]
## Rock 'n roll angels bring thyn hard rock, hallelujah
## Demons and angels, all in one have arrived
## Rock 'n roll angels bring thyn hard rock, hallelujah
## In God's creation, supernatural high
## Rock 'n roll angels bring thyn hard rock, hallelujah
## Rock 'n roll angels bring thyn hard rock, hallelujah
##
## [Outro]
## Hard rock, hallelujah!
## Hard rock, hallelujah!
##
##
## Enkel tone, to små ord: Sámiid Ednan, samisk jord
## Kom som vindpust ifra nord, ifra nord, Sámiid Ednan
##
## Kan et krav få mjukar form: Sámiid Ednan, Sameland
## Vokste sæ fra bris tel storm, bris tel storm, Sámiid Ednan
##
## Framførr tinget der dem satt, Sámiid Ednan, Sámiid Ednan
## Hørtes joiken dag og natt, Sámiid Ednan
##
## Framførr tinget der dem satt, oj...
## Hørtes joiken dag og natt, Sámiid Ednan
##
## Joik har større kraft enn krutt, Sámiid Ednan
## Førre en joik tar aldri slutt, oh... Sámiid Ednan
##
## Førre en joik tar aldri slutt, Sámiid Ednan
##
##
## Shackled bones, around all these thrones come screaming.
## Behind a wall of ego, there is a hidden hero in us, in us,

```


in all of us.
 ## You make me feel like
 ## Oh oh oh
 ## You make me, you make me feel like, yeah
 ## Oh oh oh
 ## You make me feel like
 ## Oooh
 ## You make me feel alive
 ## You make me feel like I'm alive again, alive again
 ## Only you can make me feel alive again, alive again.
 ## I'm like a little matchbox.
 ## All I need is a little spark.
 ## To light this whole world up, prepare for fire shots.
 ## Only you can make me feel alive again, alive again.
 ## Sticks and stones, might break all I own
 ## come beatin', beatin', beatin'.
 ## Behind a wall of ego, there is a hidden hero in us.
 ## Oh oh oh
 ## You make me feel like ...
 ## You make me feel like I'm alive again, alive again
 ## Only you can make me feel alive again, alive again.
 ## I'm like a little matchbox. All I need is a little spark.
 ## To light this whole world up, prepare for fire shots.
 ## Only you can make me feel alive again, alive again.
 ## You make me feel like I'm alive again, alive again
 ## Only you can make me feel alive again, alive again.
 ## It's overthrown
 ## And I am on my own alive again.
 ##
 ##
 ## Teardrops in my eyes, a dance of glistening rain
 ## Falling from the sky, a carpet of the stars
 ## Clouds of tenderness are fleeing in the light
 ## Now it's here, the light for a thousand years
 ##
 ## I tremble in your eyes, I tremble in your palms
 ## Lie inside you deep, in dreams of yours I sleep
 ## Caress you with my eyes and love you with my hands
 ## Yes, I feel the light for a thousand years
 ##
 ## I know you are my love
 ## A touch, a kiss, your heart is mine
 ## I know the light is always here
 ## We kiss each other in its shine
 ##
 ## I tremble in your eyes, I tremble in your palms
 ## Lie inside you deep, in dreams of yours I sleep
 ## Caress you with my eyes and love you with my hands
 ## Yes, I feel the light for a thousand years
 ##
 ## I know you are my love
 ## A touch, a kiss, your heart is mine
 ## I know the light is always here
 ## We kiss each other in its shine

Translation

```
songs %>%  
  sample_n(5) %>%  
  pull(translation) %>%  
  cat(sep = "\n\n\n")
```

```
## I wish I were an astronaut  
## Who looks down to the earth from up there  
## Because all those, who already were up there  
## They all say it's wonderful Here you'll feel time slower passing by  
## Silence and peace that we miss so much  
## And here today it could also be this way  
## We're small, but not necessarily bad No borders, no flags  
## From up there, the world is just beautiful  
## No countries, no nations  
## No wars can be seen from up there Everyone should go there at least once  
## To realise this struggle doesn't make sense  
## Perhaps it will take our stupid megalomania  
## And instead of talking, we begin to love No borders, no flags  
## No stupid quarrels, no different races  
## No wars, no states  
## No wars can be seen from up there Unlimited (Unlimited)  
## Peace without flags (Peace without flags)  
## From the height, borders can't be seen anymore  
## Unlimited (Unlimited), unusually  
## Without quarrels, explosions, rage and wars  
## From up there the world is just beautiful  
##  
##  
## English  
##  
##  
## English  
##  
##  
## You  
## I fell asleep, listening to your voice  
## Forgetting my life in your niche, in your niche  
## I woke up, I have found your voice back  
## And I knew how to love, tell me why, tell me why  
## You are my only love, you're my existence  
## You have given me love, you have given me the chance  
## To understand today that I lived for you  
## To know that my life doesn't exist without you  
## Since you're going to steal the memories of my childhood  
## Let me keep them just the time, just the time  
## To see again my adolescence, running in the forests  
## And letting the wind singing for you, singing for you  
## You are my only love, you're my existence  
## You gave me love, you gave me the chance  
## To understand today that I lived for you  
## To know that my life doesn't exist without you
```

```
##
##
## English
```

Place

```
songs %>% count(place)
```

```
## # A tibble: 27 x 2
##   place     n
##   <chr> <int>
## 1 -       309
## 2 1         68
## 3 10         66
## 4 11         58
## 5 12         60
## 6 13         70
## 7 14         59
## 8 15         53
## 9 16         56
## 10 17         48
## # i 17 more rows
```

```
length(unique(songs$year))
```

```
## [1] 66
```

```
songs %>%
  count(year, place) %>%
  filter(place == "1") %>%
  arrange(-n)
```

```
## # A tibble: 65 x 3
##   year place     n
##   <chr> <chr> <int>
## 1 1969 1         4
## 2 1956 1         1
## 3 1957 1         1
## 4 1958 1         1
## 5 1959 1         1
## 6 1960 1         1
## 7 1961 1         1
## 8 1962 1         1
## 9 1963 1         1
## 10 1964 1         1
## # i 55 more rows
```

According to Wikipedia, there were four winners in 1969:

https://en.wikipedia.org/wiki/Eurovision_Song_Contest

On only one occasion have multiple winners been declared in a single contest: in 1969, four countries finished the contest with an equal number of votes and due to the lack of a tie-break rule at the time, all four countries were declared winners.[7][157]

Rule 6: You May Need to Go Back

Our original parse function just grabbed `translation` as is. After looking at `translation`, we can see that this variable says `English` if the original song lyrics are in English. `English`.

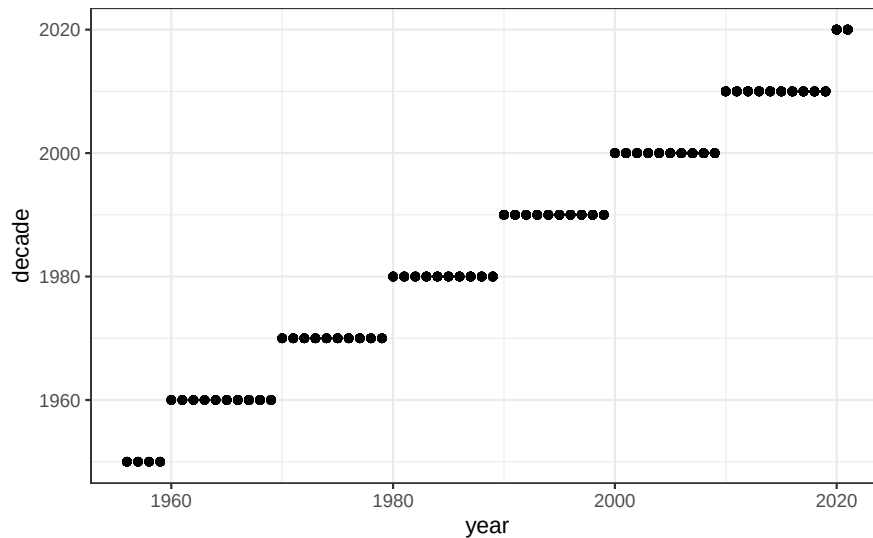
Rule 7: Create New Columns?

Decade

songs

```
## # A tibble: 1,644 x 7
##   country      artist      title    year lyrics translation place
##   <chr>      <chr>      <chr>    <chr> <chr> <chr>      <chr>
## 1 Netherlands Jetty Paerl    De voge~ 1956 "De v~ "The birds~ -
## 2 Switzerland Lys Assia     Das alt~ 1956 "Das ~ "The old c~ -
## 3 Belgium     Fud Leclerc   Messieu~ 1956 "Mess~ "Ye drowne~ -
## 4 Germany (West) Walter Andreas Schwarz Im Wart~ 1956 "Es g~ "There is ~ -
## 5 France      Mathé Altéry  Le temp~ 1956 "Chan~ "Sing, car~ -
## 6 Luxembourg  Michèle Arnaud Ne croi~ 1956 "Si o~ "If they t~ -
## 7 Italy        Franca Raimondi Aprite ~ 1956 "La p~ "The first~ -
## 8 Netherlands Corry Brokken Voorgoe~ 1956 "Voor~ "Over for ~ -
## 9 Switzerland Lys Assia (2) Refrain  1956 "[Int~ "Chorus of~ 1
## 10 Belgium    Mony Marc     Le plus~ 1956 "Les ~ "The bells~ -
## # i 1,634 more rows
```

```
songs <-
  songs %>%
  mutate(
    year = parse_number(year),
    decade = year - year %% 10
  )
songs %>%
  ggplot(aes(year, decade)) +
  geom_point()
```



Lyrics

```
songs %>%
  count(translation) %>%
  arrange(-n)
```

```
## # A tibble: 946 x 2
##   translation                                n
##   <chr>                                     <int>
## 1 "English"                                694
## 2 ""                                         2
## 3 "Day is turning to evening, is gone, soon forgotten\nBut what we gave ~ 2
## 4 "God, god, god, god they don't call you\nCandles, candles, candles, ca~ 2
## 5 "Look into my eyes now,\nI'm not a loser, know it!\nI don't surrender~ 2
## 6 "Only champions!\nOnly champions! They fly high1 and jump far2.\nAll t~ 2
## 7 "\nPure and white, her skin is so inviting\nLike a new, unbeaten snow\~ 1
## 8 "\"Come with me\"", said my friend\nThe first silent winter snow is fal~ 1
## 9 "\"This is the Telephone Information Service, hello\"\n\n\"Yes, hello, t~ 1
## 10 "\"We are all moving around the old circle\"1,\nSinging in this sweet ~ 1
## # i 936 more rows
```

```
songs <-
  songs %>%
  mutate(
    english_lyrics = ifelse(
      translation == "English",
      lyrics, translation
    )
  )
```

Rule 7: Save the Data and Write it Up

It's critically important to make detailed notes about any changes you have made when cleaning the data.

```
filenamepath <- glue::glue("./data/{lubridate::today()}-eurovision.rds")  
filenamepath
```

```
## ./data/2025-02-17-eurovision.rds
```

```
write_rds(songs,filenamepath)
```